How to use clustering shinyapp

Bingtian Ye
btye@bu.edu

Content

- 1. Preface
- 2. How to open clustering shinyapp
- 3. How to use clustering shinyapp
- 4. Download from github (for people not in Waxman Lab)
- 5. Acknowledge

1.Preface

The Clustering Shiny App was developed by Bingtian Ye (Waxman Lab). Built upon the Seurat framework, it incorporates modules for normalization, scaling, PCA, graph construction, Harmony integration, clustering, and subsetting. It also features corresponding UMAP plots, heatmaps, and dot plots.

Github link:
https://github.com/Ye1203/Clustering_shinyapp

SCC Folder Path:

/projectnb/wax-es/00_shinyapp/Clustering

Contact e-mail address:

btye@bu.edu, biangtian@icloud.com

Publication method section template path:

/projectnb/wax-es/00_shinyapp/00_Tutorial_Description/Clustering/More information of Clustering (publication method section).docx

2. How to open clustering shinyapp

If you do not have access to wax-es project in BU SCC, you can skip to Step 4 and download from GitHub.

If you do not run the code on BU SCC, you cannot use the function “Search on SCC background”.

If you have access to wax-es project, please follow following step to open clustering shinyapp.

a. Apply a Rstudio server and launch it

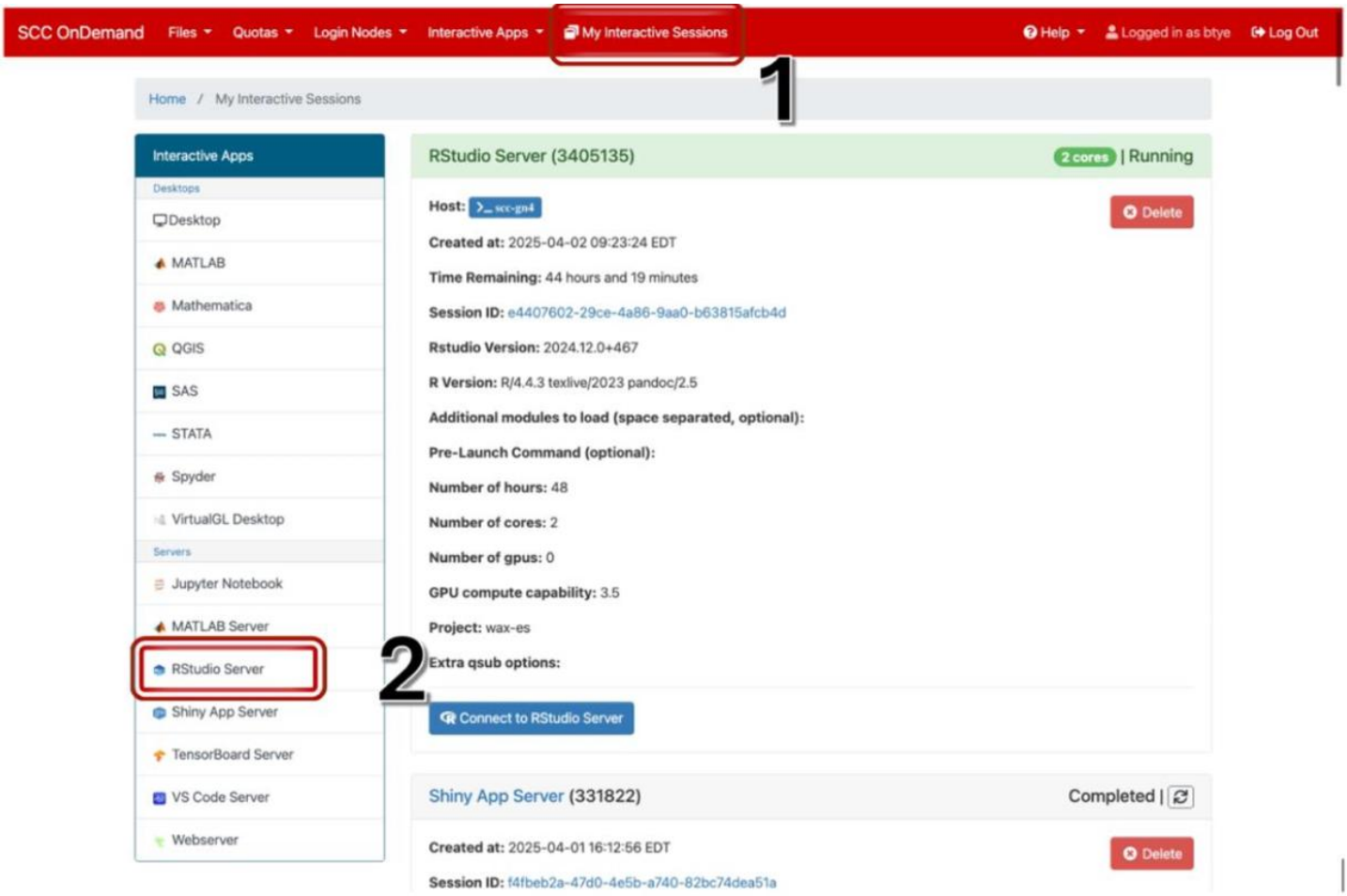
b. Open the project and launch the environment

c. Launch shinyapp

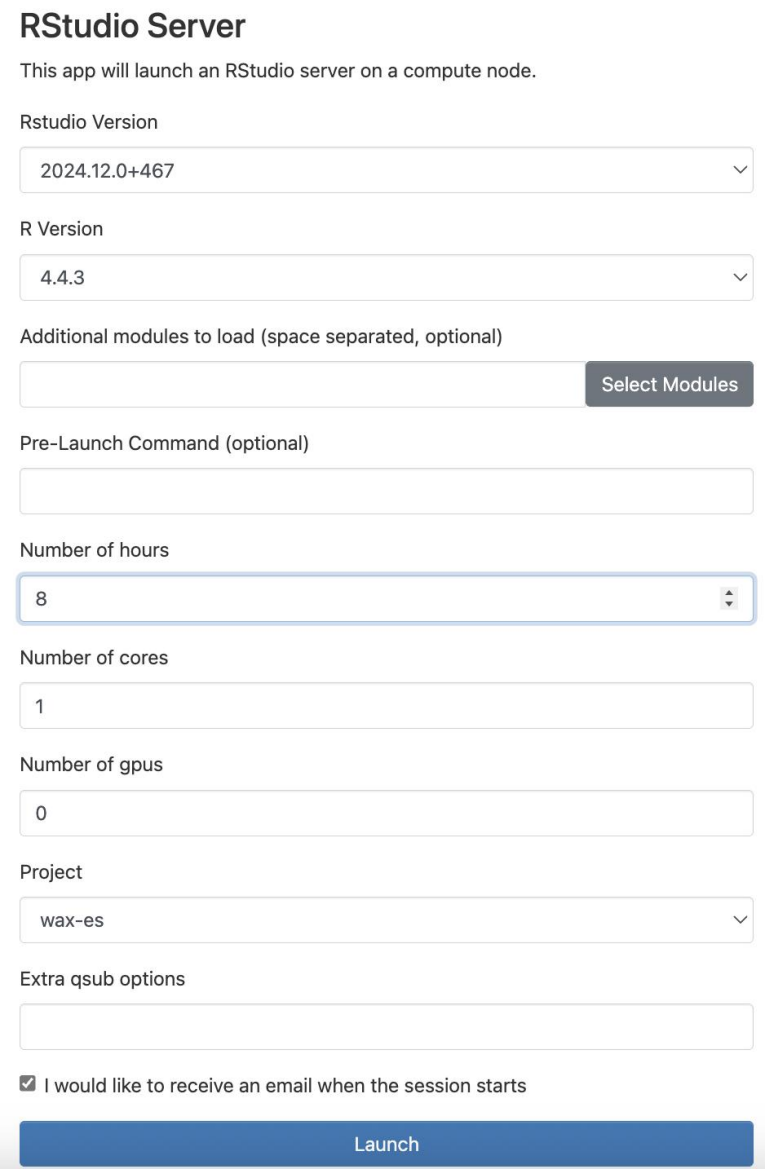
a. Apply a Rstudio server and launch it

Open scc and click as following:

My Interactive Sessions -> Rstudio Server



Setting Rstudio Server



R Version: must be **4.4.3**

Number of hours: 8 is enough, you can apply more if you need

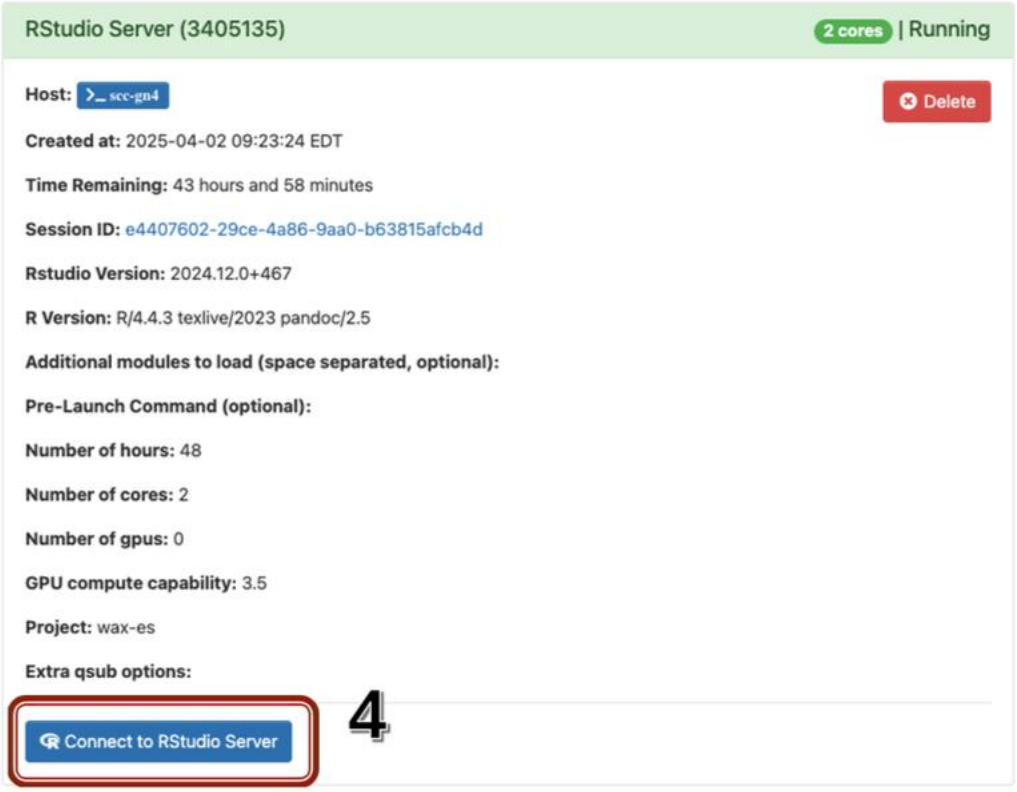
Number of cores: 1

Number of gpus: 0

Project: any project you can access
After all, click Launch

Connect to Rstudio Server

After few minutes the Rstudio server is running, click “Connect to Rstudio Server” to open Rstudio.

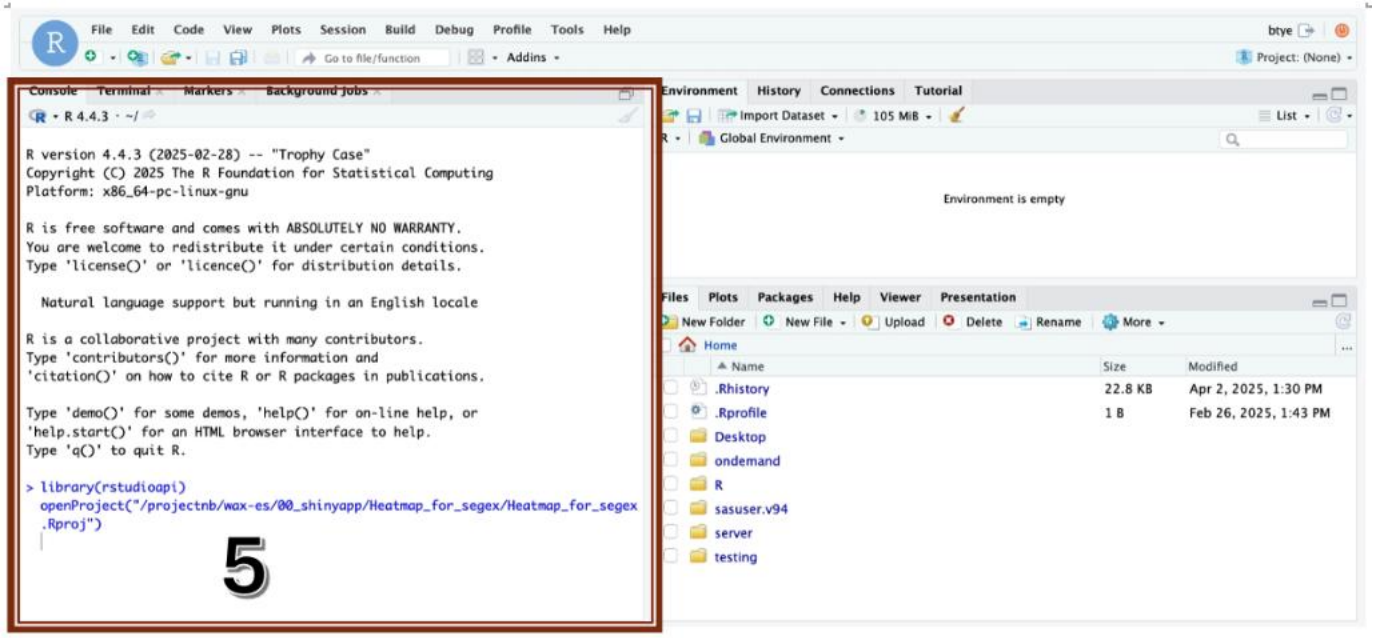


b. Open the project and launch the environment

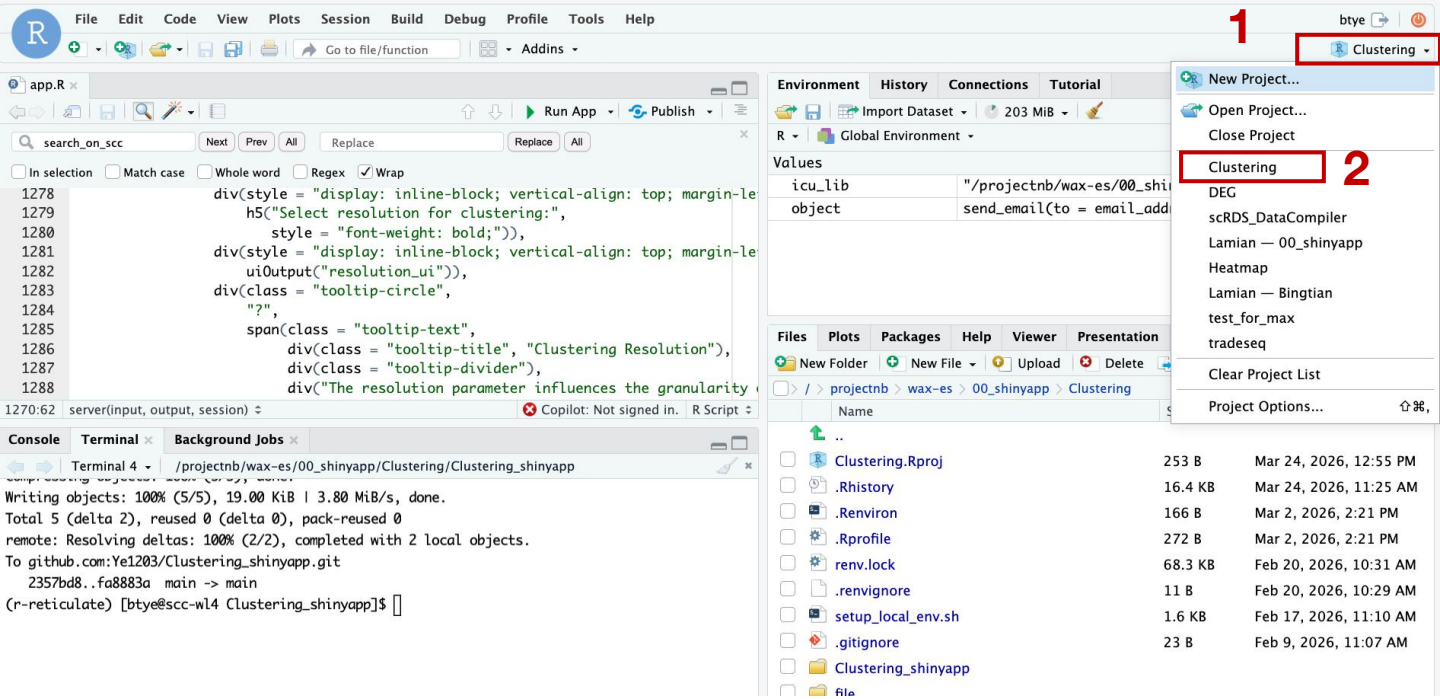
The environment is build by renv. It is recommend to use R Project to launch the environment

Copy the following red text, then paste into the “Console” tab (blue test below), and then hit “Enter”.

library(rstudioapi)
openProject("/projectnb/wax-es/00_shinyapp/Clustering/Clustering.Rproj")



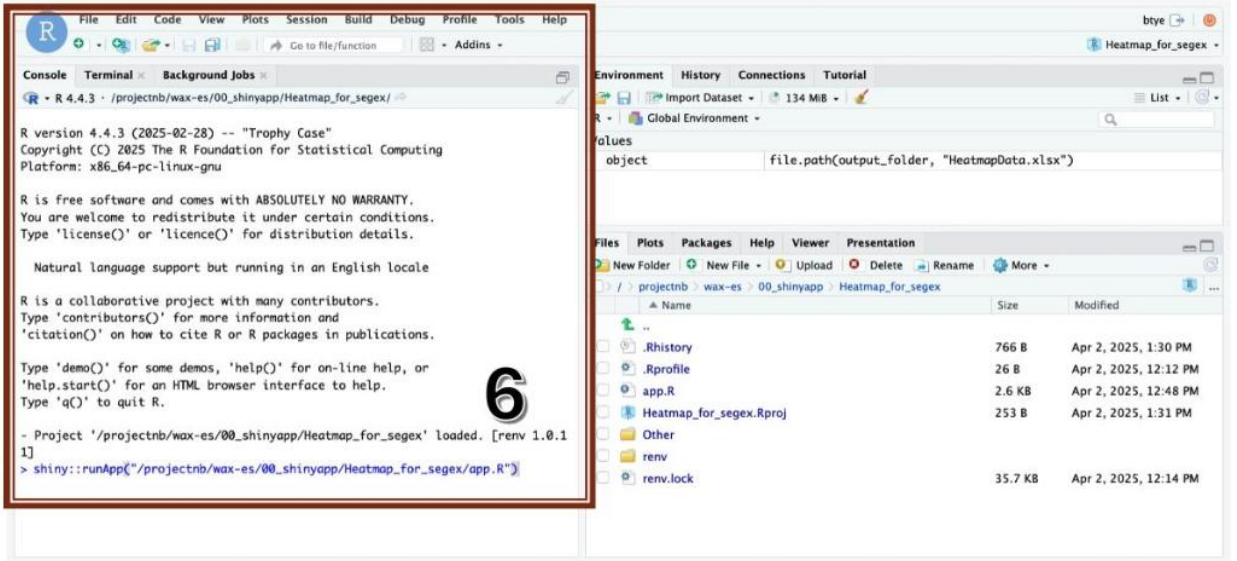
If you already launch the project previously, you can transfer the project using Rstudio.



c. Launch shinyapp

Copy the following red text, then paste into the "Console" tab (blue test below), and then hit "Enter".

shiny::runApp("/projectnb/wax-es/00_shinyapp/Clustering/Clustering_shinyapp/app.R")



3. How to use clustering shinyapp

This document will use the follow RDS file as example:

/projectnb/wax-es/00_shinyapp/00_Tutorial_Description/Clustering/example_data.rds

This file is mouse liver seurat file. Which included 8 samples (G183M1, G183M2, G183M3, G183M4, G193M1, G193M2, G193M3, G193M4) in the meta.data `sample_id`

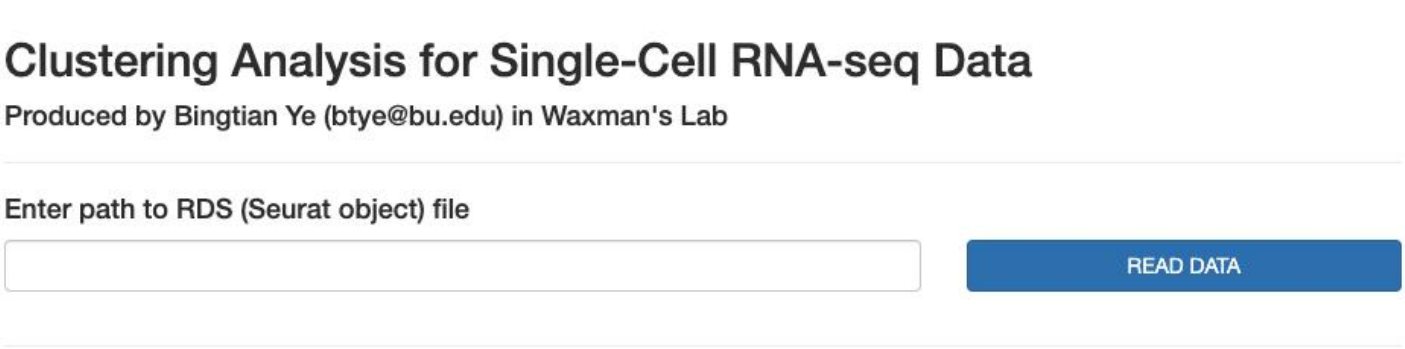
Clustering included steps:

a. Input path & SAVE DATA
b. DATA NORMALIZED
c. SELECT FEATURFE AND DATA SCALED
d. RUN PCA
e. FIND NEIGHBORS
f. FIND CLUSTERS
g. RENAME AND SUBSET

SAVE DATA

a. Input path & SAVE DATA

Input path



Copy the path from SCC or your own PC (if you run the shinyapp on your own PC). And Click `READ DATA`. The RDS file will be load to the shinyapp.

After READ Data, the ui will turn to:

You can click `RESET` to Remake all content and re-enter the new files.

b. DATA NORMALIZED

Enter path to RDS (Seurat object) file

Loaded: /projectnb/wax-es/00_shinyapp/00_Tutorial_Description/Clustering/example_data.rds

RESET

DATA NORMALIZED

Normalize Data

The section cannot be run (You must run previous step).

SELECT FEATURE AND SCALE THE DATA

This section has already been executed.

RUN PCA

This section did not execute, yet the corresponding data already exists. This typically occurs when the imported data has previously completed the relevant steps; you may either skip these sections or rerun them.

FIND NEIGHBORS

FIND CLUSTERS

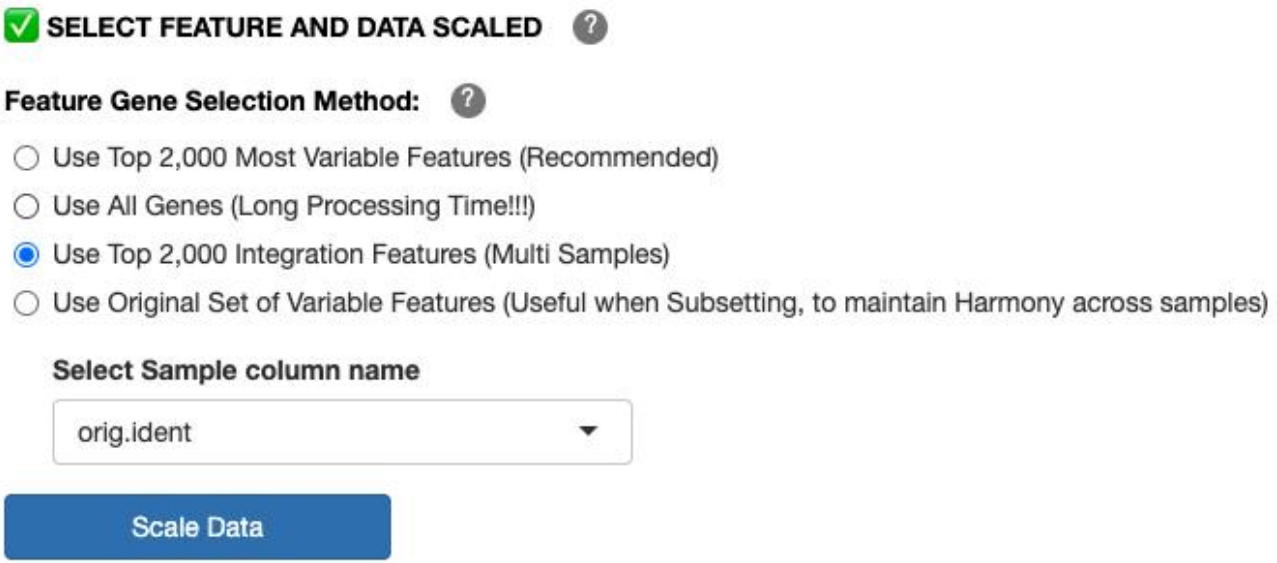
RENAME AND SUBSET

Data Normalized (Seurat::NormalizeData)

$$y_{ij} = log\left(\frac{x_{ij}}{\sum_i x_{ij}} \times 10000 + 1\right)$$

x_{ij}: Feature i's UMI in Cell j

c. SELECT FEATURFE AND DATA SCALED - 1



Select Feature and Scale Data included feature selected and data scale.

Feature scale included 4 option, which will select the feature gene for the following scale, pca.

Scale data uses Seurat::ScaleData (z-scale)

You only need to enter addition sample information (Select Sample column name) when selecting option `Use Top 2,000 Integration Features (Multi Samples)`

c. SELECT FEATURFE AND DATA SCALED - 2

Feature Gene Selection Method:

- ○ Use Top 2,000 Most Variable Features (Recommended)
- ○ Use All Genes (Long Processing Time!!!)
- ◉ Use Top 2,000 Integration Features (Multi Samples)
- ○ Use Original Set of Variable Features (Useful when Subsetting, to maintain Harmony across samples)

4 Options for select feature:

1. Use Top 2,000 Most Variable Features (Recommended)
   Use Seurat::FindVariableFeature. Based on the expression level and variance. Consider using other methods only when the results of this method are unsatisfactory.
2. Use All Genes (Long Processing Time!!!)
   Using all available features may introduce noise and consumes significant time and memory; therefore, it is not recommended.
3. Use Top 2,000 Integration Features (Multi Samples)
   Use Seurat::SelectIntegrationFeatures. Specifically designed for multiple samples, this method first calculates the features for each individual sample, and then selects the 2,000 most frequently occurring genes to serve as the features.
4. Use Original Set of Variable Features (Useful when Subsetting, to maintain Harmony across samples)
   This applies specifically to scenarios involving subsetting—for instance, when you have subsetted only a specific cell type, and these cells exhibit a very high degree of similarity. In such cases, it may be necessary to utilize the features originally computed using the entire set of cells.

d. RUN PCA

PCA Dimension Reduction (Seurat::RunPCA)

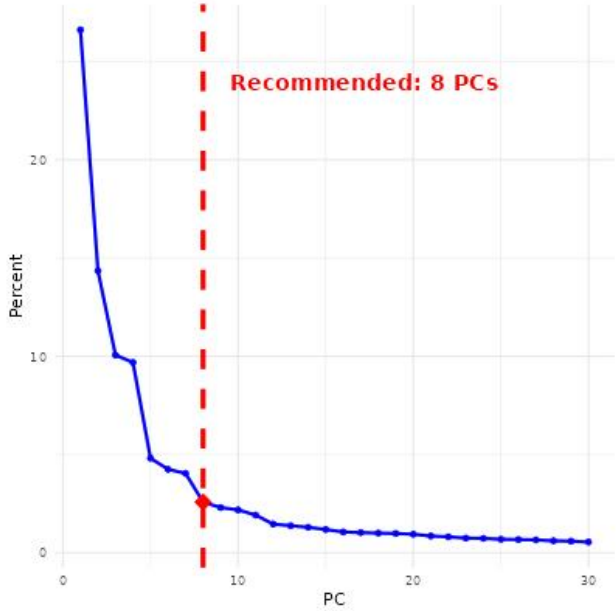
Based on Feature Gene.

The input is the number of saved PCA components.
It is recommended to keep the default setting; you should only increase the PCA count if the subsequent `Find Neighbor` step recommends a value of 30.

e. FIND NEIGHBORS - 1

✅ FIND NEIGHBORS

Elbow Plot with Recommended Dimensions:



Dimension Selection Criteria

Recommended dimensions based on:

- **Priority 1:** PCs after the elbow point, where variance drop < 0.5%.
- **Priority 2:** Cumulative variance ≥ 90%.

Elbow Point: **5 PCs**

Last variance drop < 0.5% point: **8 PCs**

Cumulative variance ≥ 90% point: **17 PCs**

Recommended dimensions: **8 PCs**

Only appear when `Harmony Integration (if you have multi sample in the data)`

Select the number of dimensions (PCs) to use for finding neighbors: 8

☑ **Harmony Integration (If you have multi samples in the data)**

Select metadata column for Harmony integration (sample identification): sample_id

Preview values: G183M1, G183M2, G183M3, G183M4, G193M1, G193M2, G193M3, G193M4

Find Neighbors | Check Harmony Result

Only appear after `Find Neighbors`

e. FIND NEIGHBORS - 2

The Default of `Select the number of dimensions (PCs) to use for finding neighbors` is recommened dimensions. You can edit the value as you want.

If you are using multisampling, it is strongly recommended to use the harmony integration feature to remove batch effect.

1. Check `Harmony Integration (If you have multi samples in the data` to remove batch effect.
2. Select metadata column for Harmony integration (sample identification)
3. Check the sample identification by take a look of `Preview values`
4. Click `Find Neighbors`

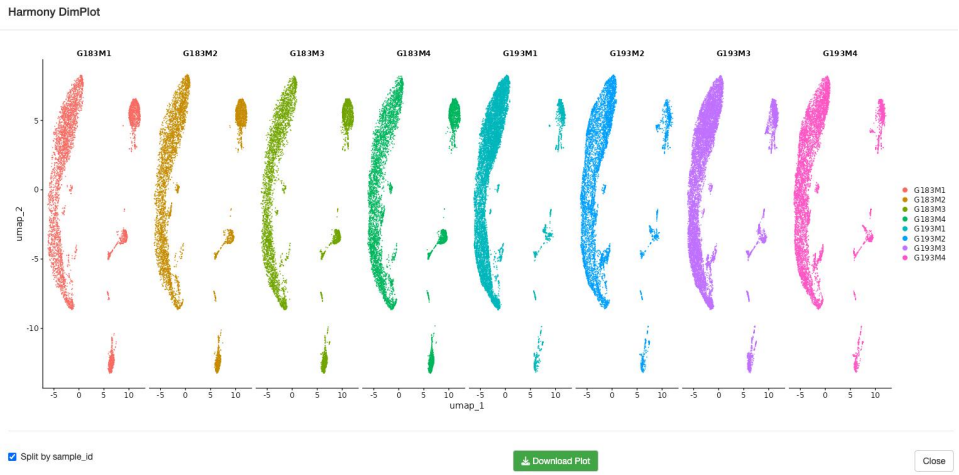
If you RDS file only includes one sample, don't use harmony integration but click `Find Neighbors` Directly

After `Find Neighbors`, you can click `Check Harmony Result` to check and download Harmony result (next slide)

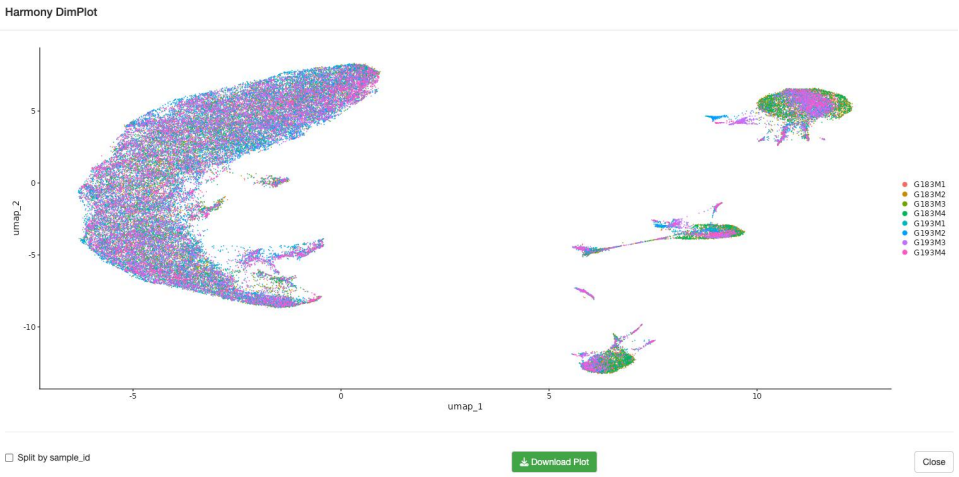
e. FIND NEIGHBORS - 3

Check Harmony Result

After click `Check Harmony Result`, you will open following ui:

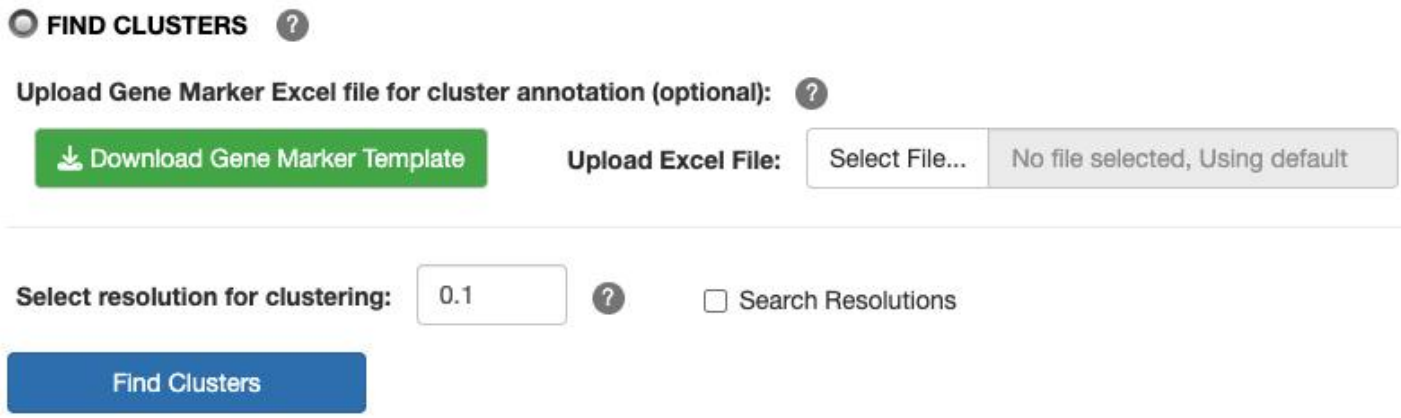


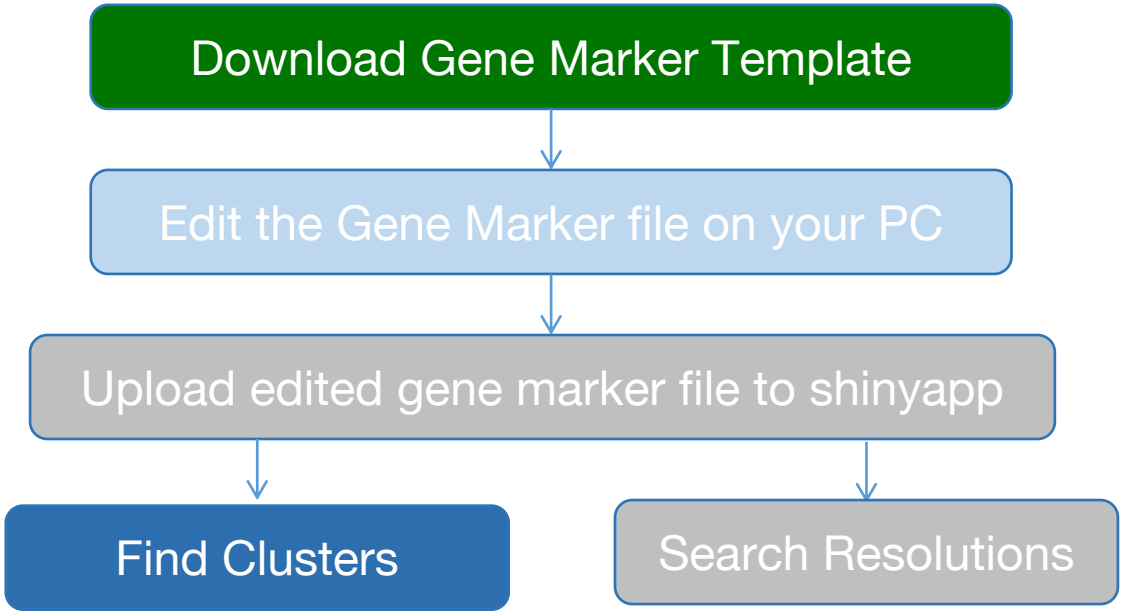
Check `Split by sample_id` or not to see the combined sample_id plot or individual sample_id plot.



Click `Download Plot` to download current plot to your PC

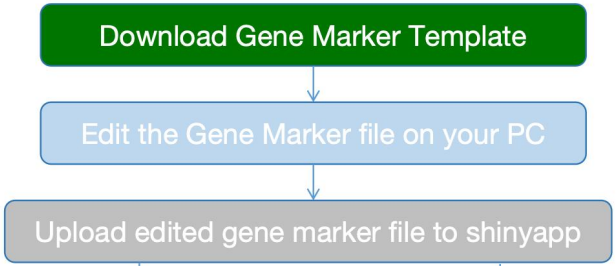
f. FIND CLUSTERS - 1





`Search Resolutions` allows you to calculate and verify results at different resolutions.

f. FIND CLUSTERS - 2



After Click `Download Gene Marker Template`. A excel file called `Gene_Markers.xlsx` will download to your PC.

	A	B	C	D	E	F
1	Cell Type A	Cell Type B	Cell Type C	Cell Type D	Cell Type E	Cell Type F
2	A1	B1	C1	D1	E1	F1
3	A2	B2	C2	D2	E2	F2
4	A3	B3	C3	D3	E3	F3
5	A4	B4		D4	E4	F4
6	A5	B5		D5	E5	F5
7	A6	B6		D6	E6	F6
8	A7	B7			E7	F7
9	A8	B8			E8	F8
10	A9	B9				F9
11	A10	B10				F10
12	A11	B11				F11
13		B12				F12
14		B13				F13
15		B14				F14
16						

HEATMAP DOTPLOT1 DOTPLOT2 +

The `Gene_Markers.xlsx file included 3 sheets. Each sheet includes column name (Celltype) and value (Gene/Feature)

`HEATMAP` is used for identify cluster and heatmap. DOTPLOT is used for two dotplot

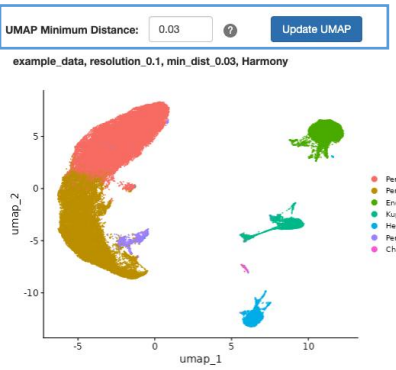
After upload your edited file, shinyapp will check your uploaded file automatically.

This includes checking whether the uploaded marker file contains markers that are absent from the RDS file, or if duplicate markers appear in the DOTPLOT. The former will be automatically ignored, whereas for the latter, you must remove the duplicate genes before re-uploading.

f. FIND CLUSTERS - 3

Find Clusters

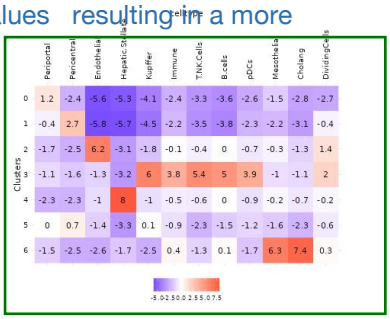
Input the resolution (Higher resolution tend tomore cluster) and click `Find Clusters`, you will get following plot.



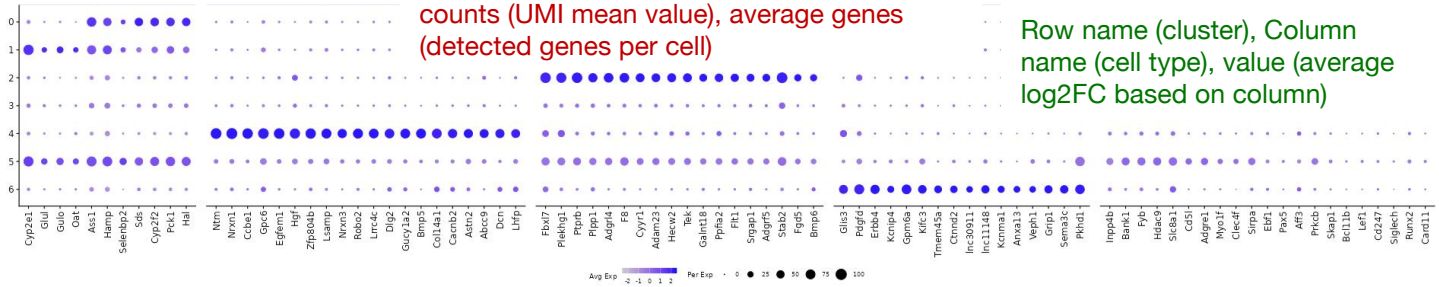
`UMAP Minimum Distance` is used to adjust the degree of point dispersion in a UMAP plot; it ranges from 0 to 1, with higher values resulting in a more dispersed plot.

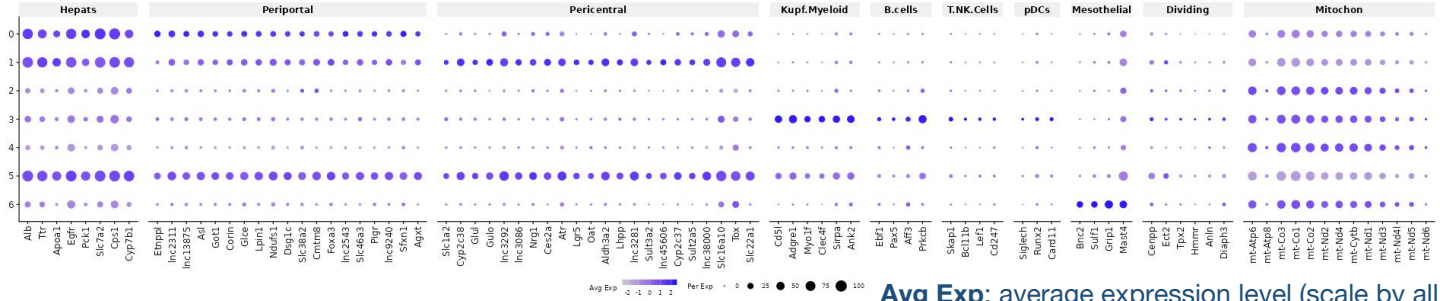
cluster	ncell	pct	avg.counts	avg.genes
0	26248	38.58	2256	1296
1	23491	34.53	4984	2215
2	8173	12.01	1076	742
3	4430	6.51	1332	883
4	3829	5.63	1333	797
5	1293	1.90	13773	3841
6	568	0.83	1250	813
total	68032	99.99	3155	1536

data table. Inclueded differnet cluster's cell number, percent of cell number, average counts (UMI mean value), average genes (detected genes per cell)



Row name (cluster), Column name (cell type), value (average log2FC based on column)





Avg Exp: average expression level (scale by all marker - two dotplot)
Per Exp: Non-zero gene percentage

Download All visualization result (next slide)

f. FIND CLUSTERS - 4

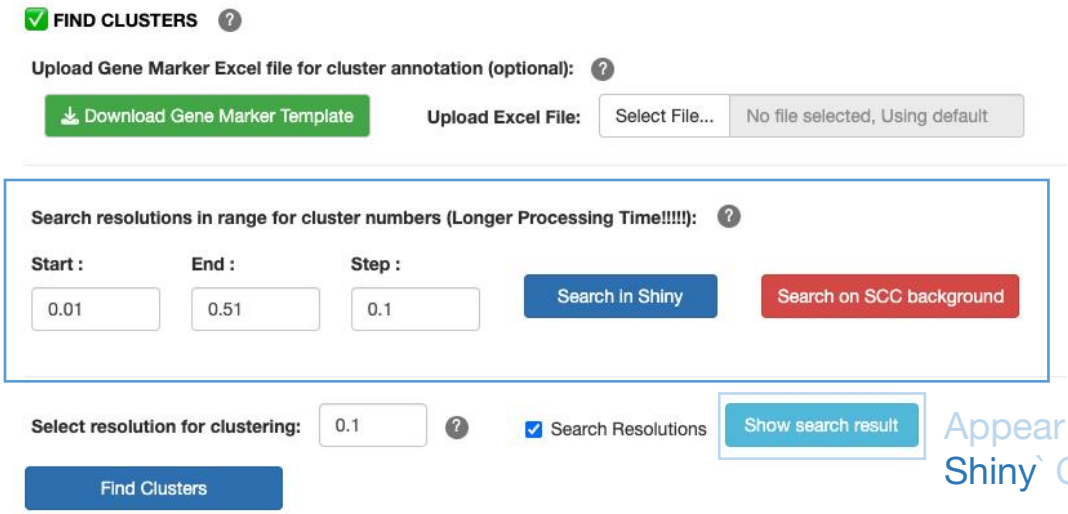
Find cluster download files:

1. Cluster_UMAP
   umap plot pdf file
2. Cluster_Heatmap
   heatmap plot pdf file
3. Cluster_Dotplot1, Cluster_Dotplot2
   dotplot pdf file
4. Cluster_Combined
   combined umap, data_table, heatmap and dotplot pdf file
5. Cluster_data
   All plot's data excel file
   a. table_data: from table
   b. heatmap_data: log2FC in heatmap
   c. dotplot_avg_exp: dotplot's average expression level
   d. dotplot_pct_exp: dotplot's percentage of expression

f. FIND CLUSTERS - 5

Search Resolutions

Search resolutions (Optional). allows you to calculate and verify results at different resolutions (Check resolution space).



Appear after check `Search Resolutions`

Appear after `Search in Shiny` Completed

1. Check `Search Resolutions`.
2. Input the `Start` (from 0.0001 to 5), `End` (from 0.0001 to 5) and `Step` (from 1e-10 to 1) value. (e.g. The values shown in the figure are analyzed at resolutions of 0.01, 0.11, 0.21, 0.31, 0.41, and 0.51).
3. Start Analysis.
   Option1. Only get the different resolutions' number of cluster. -> `Search in Shiny`
   Option2. Get different Resolutions' visualization result -> `Search on SCC background`

f. FIND CLUSTERS - 6

Search Resolutions

Search in Shiny

Executing the `Search in Shiny` will generates a table displaying the number of clusters at various resolutions.

This function runs directly on the SCC cluster; you must wait for all analysis tasks to complete, as the entire process may take some time.

Once the analysis is finished, if you wish to review the table again, please click "Show search result".

Resolution Search Results

Resolution	Number_of_Clusters
0.01	4.00
0.11	7.00
0.21	10.00
0.31	13.00
0.41	17.00
0.51	18.00

Dismiss

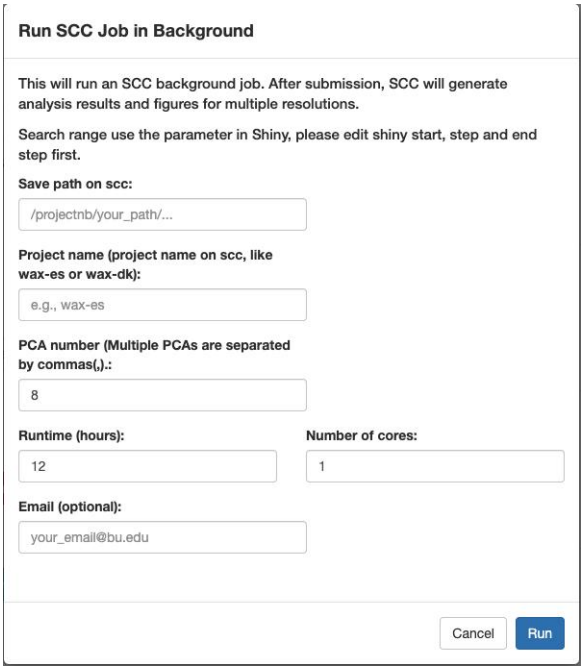
f. FIND CLUSTERS - 7

Search Resolutions

Search on SCC background

Search on the SCC background is a powerful tool designed for users who have access to SCC or are familiar with using `qsub`. This feature automatically generates all required inputs, R scripts, and submission commands, and runs the jobs in the background. All visualization results will be saved to the specified output path, allowing you to close the Shiny app or continue with other analyses (e.g., clustering) without interruption.

If you plan to explore a large number of resolutions, we strongly recommend using this function. After Click `Search on SCC background`, you will open following window:



Save path: Output folder on SCC where all results will be saved
Project name: Your SCC project name (used as `-P` in `qsub`)
PCA numbers: One or more values, separated by commas
Runtime & cores: Use defaults or specify a runtime and number of cores
Email (optional): Receive a notification when the job complete

After clicking "Run", please wait patiently for the analysis to start, and then close the window.

Upon completion of the run, several files will be generated in the save path. (next slide)

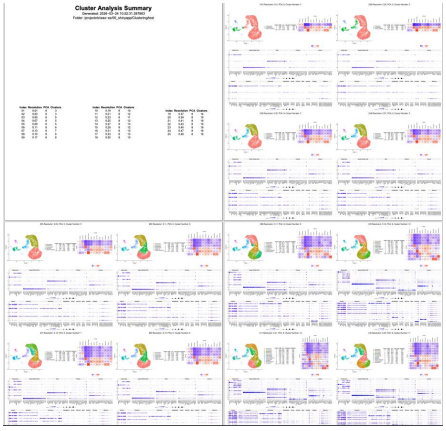
f. FIND CLUSTERS - 8

Search Resolutions

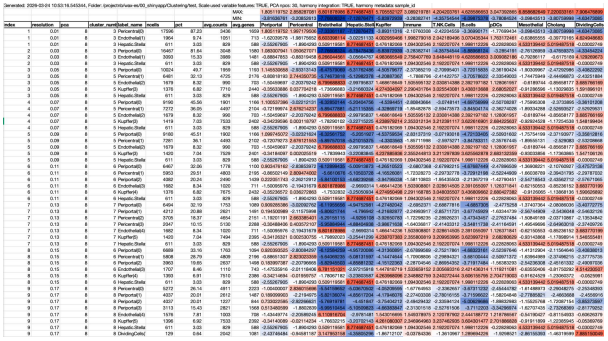
Search on SCC background

Output:

1. different rds file (original, different pca number)
2. Gene Marker file (Gene_Markers.xlsx)
3. submission command file (launch_job.qsub)
4. R code (harmony_job_script.R, scc_job_script.R)
5. command information (command_info.txt)
6. log file (cluster_job.log)
7. Visualization result folder
   a. Different resolutions' visualization plot (All_Resolutions_Combined.pdf), included number of cluster table and plot.
   b. Different resolutions' data (Resolution_ClusterNumber_Summary.xlsx), included expression level data and heatmap data



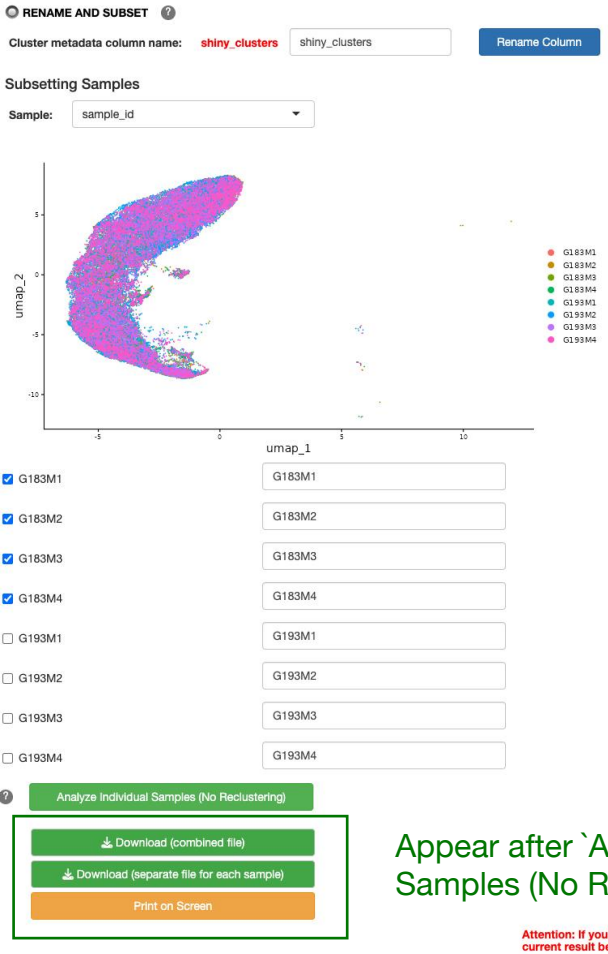
a. Different resolutions' visualization plot

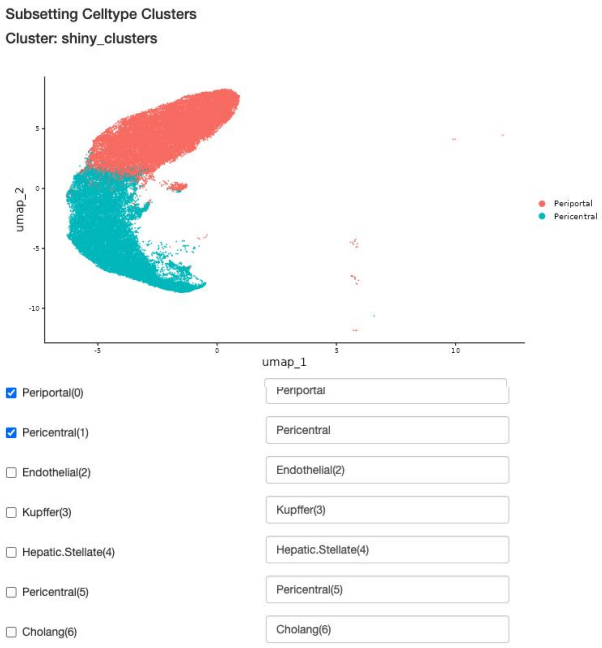


b. Different resolutions' data

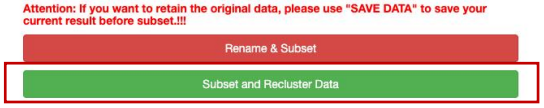
In this way, you can identify the resolution value you want; then, input this value and run `Find Clusters` (see "Find clusters - 3").

g. RENAME AND SUBSET - 1



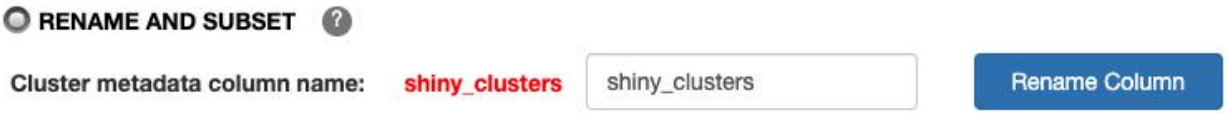


Appear after `Analyze Individual Samples (No Reclustering)`



Appear after `Rename and Subset`

g. RENAME AND SUBSET - 2



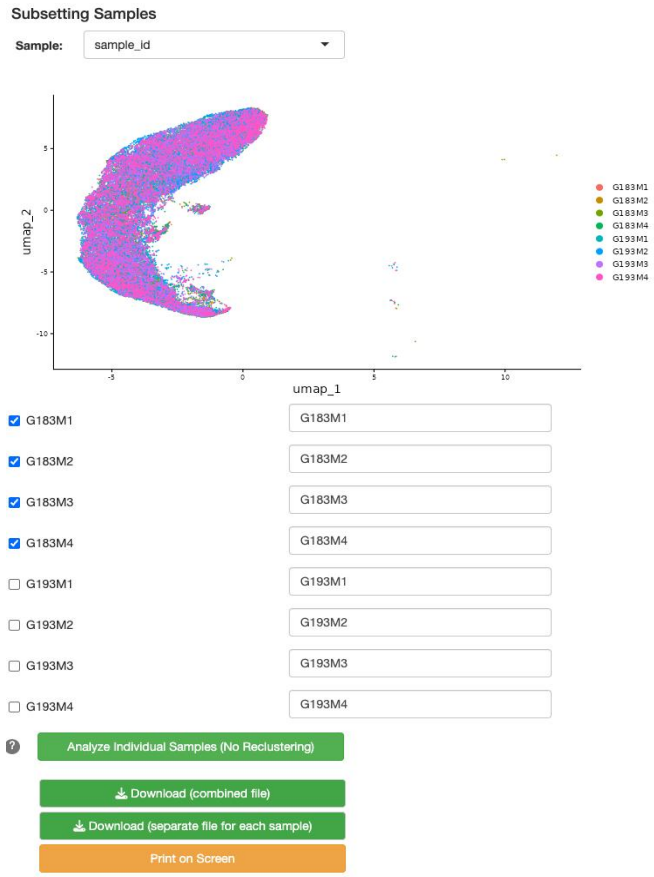
Rename the meta data column name for the clustering (Optional)
After `Find Cluster`, the clustering information will save as `shiny_clusters` in the meta.data as default. You can change the name as you want.

Subset and Rename
You can filter out unwanted cells by deselecting the specific samples or cell types you wish to exclude; you also have the option to rename these categories. For instance, in this example, I retained only "Periportal(0)" and "Pericentral(1)," renaming them simply to "Periportal" and "Pericentral."

Once you have finished making your modifications, click `Rename & Subset`. This action generates a temporary file—meaning that even if you initially removed certain cells, you can still add them back later—containing your revised data, and the UMAP plot will automatically update to reflect these changes.

At this stage, you can click "Subset and Recluster Data" (note that this action is irreversible) to analyze the resulting subset of data. This process resets the analysis pipeline, automatically replacing your original input with the subsetted data, thereby allowing you to proceed with downstream analyses such as scaling, PCA, and so forth. It is highly recommend to save the file (view SAVE DATA section) before click "Subset and Recluster Data"

g. RENAME AND SUBSET - 3



Analysis Individual Samples (No clustering)

This function allows you to view distribution differences across different samples.

Select the samples you need (for example, in this case, I only want to analyze sample G183), then click the `Analysis Individual Samples (No clustering)`.

Afterwards, three options will appear:

1. Download (combined file)
   A PDF file containing the visualization results for all samples, along with an Excel file containing the data, will be downloaded.
2. Download (separate for each sample)
   Includes a separate PDF visualization report for each sample, as well as an Excel file containing the data.
3. Print on Screen
   Display visualization results directly within Shiny (suitable only for small sample sizes).

SAVE DATA

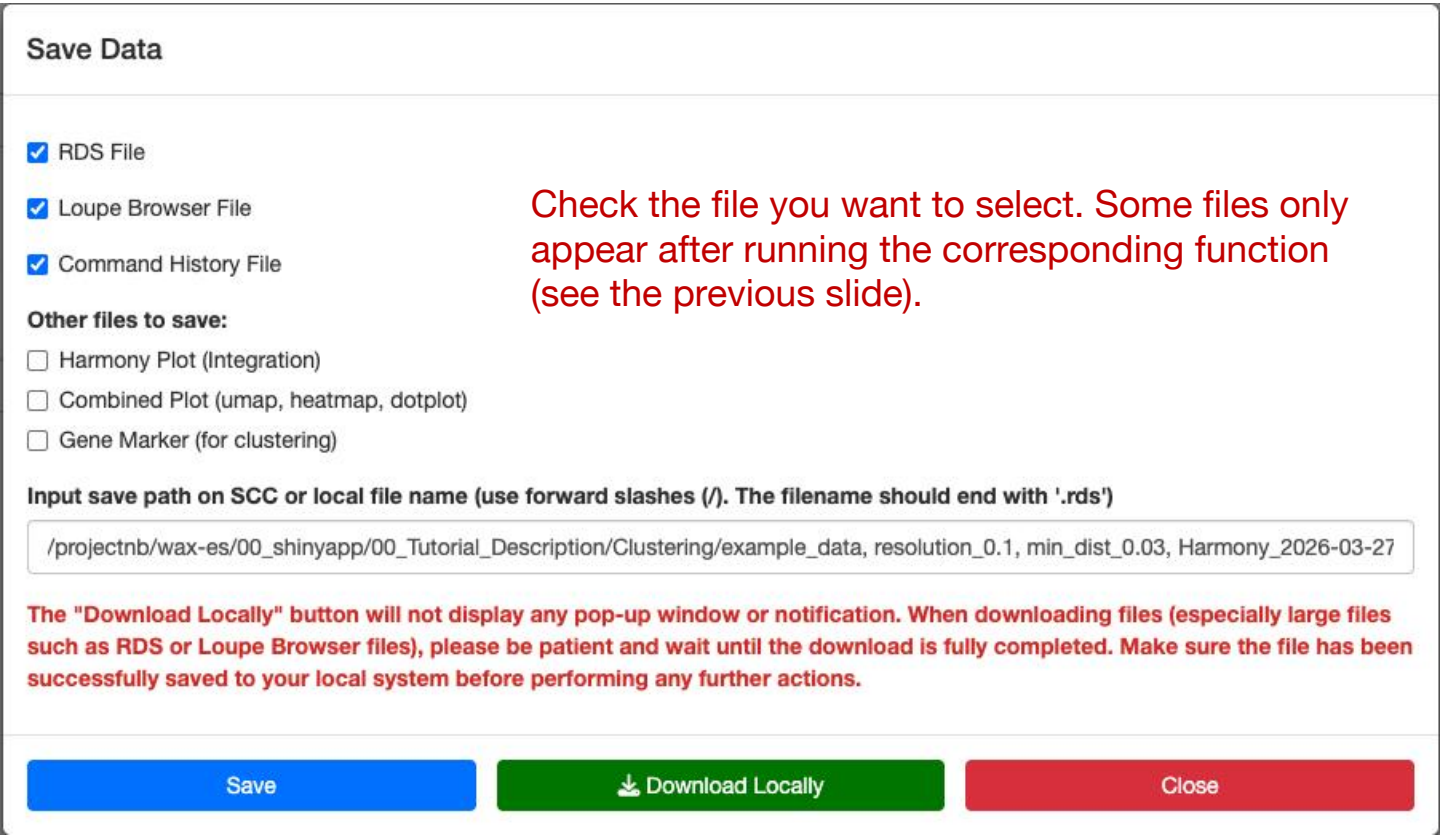
There is a `SAVE DATA` button In the bottom left corner (After read data).
It will allow you to download your current result to your own PC or save it to SCC. For example, if you have completed the PCA, the download RDS file will included the PCA result.
Please remind, the `RENAME AND SUBSET` section will generate temporary file which included subset and rename result after you click `Rename & Subset`. `SAVE DATA` will check whether there is a temporary file first. If there is temporary file, `SAVE DATA` will save it first.

There are 6 differents file you can select to save:

	RDS File	RDS File (.rds), seurat obj.
	Loupe Browser File	Loupe Browser File (.cloupe). For Loupe Browser application.
	Command History File	Included all command you use in clustering shinyapp.
Only appear after hamrony	Harmony Plot (integration)	Umap plot, separate by sample.
Only appear after find cluster	Combined Plot (umap, heatmap, dotplot)	Included umap, heatmap and dotplot and one combined plot.
	Gene Marker (for clustering	The Gene Marker you used for clustering analysis.

After Click “SAVE DATA”, you will see a window (next slide) to help you save your file.

SAVE DATA



Check the file you want to select. Some files only appear after running the corresponding function (see the previous slide).

- Regardless of whether you intend to save the RDS file, the save path must end with the `.rds` extension. The save path specifies a location on the SCC; if you choose to Download Locally, a `.zip` archive will be generated, the name of which will match the filename specified in the save path.
- Click `Save` to save your selected file to SCC.
- Click `Download Locally` to download the file to your PC.
- The `Save` function displays a pop-up notification indicating whether the process is still running; however, `Download Locally` does not. Therefore, if you choose to download the file to your local, please ensure that the resulting .zip file is present locally before proceeding with any further actions.

4. Download from github

We also upload the code to github for people cannot use SCC.

```
git clone https://github.com/Ye1203/Clustering_shinyapp.git
```

You can use renv.lock to rebuild the environment (recommend) or download all packages by yourself.

Some code may need change:
app.R. line 1575:

```
source(file.path(getwd(), "scc_job_submission.R"))
```

5. Acknowledge

This tool was developed by Bingtian Ye at Waxman's Lab, Boston University.

If you use this tool in your research, please acknowledge it by citing or mentioning it in your publication.

For questions, feedback, or bug reports, please contact:
btye@bu.edu or biangtian@icloud.com